

Software Engineering Laboratory

Lab Sheet – 4

Date	21 August 2026
Course	Software Engineering Laboratory
Experiment No.	4
Title	Programming Assignment – Real-World Software Applications
Batch	IMG
Instructor	Dr. Dheeraj Kodati, Assistant Professor

1. Aim

To design, implement, and test software applications for real-world problems using appropriate programming concepts and software engineering practices.

2. Learning Objectives

Students will be able to:

- Analyze a real-world software problem.
- Design a solution before coding.
- Implement a functional software application.
- Apply input validation and error handling.
- Test programs using different test scenarios.
- Demonstrate and evaluate the developed application.

3. Task 1 – Hospital Appointment Management System

Develop a console-based **Hospital Appointment Management System** for a hospital. The program should provide a menu-driven interface.

The system should allow the user to:

1. Add a patient.
2. Display all patients.
3. Search for a patient.
4. Book an appointment.
5. Cancel an appointment.
6. Update appointment details.
7. Display appointments for a doctor.
8. Display an appointment report.
9. Exit the application.

Patient Details

Maintain at least: Patient ID, Patient Name, Doctor Name, Department, Appointment Date, and Appointment

Status. **Required Validation**

The program must handle: Duplicate Patient ID; Patient ID not found; Empty patient name; Empty doctor name;

Invalid appointment date; Appointment already booked; Cancelling a non-existing appointment; Invalid menu choice; and Invalid numerical input.

4. Task 2 – Restaurant Order Management System

Develop a console-based **Restaurant Order Management System**.

The system should allow the user to:

1. Add a food item.
2. Display all food items.
3. Search for a food item.
4. Add a food item to an order.
5. Remove a food item from an order.
6. Update food quantity.
7. Display the customer order.
8. Calculate the bill.
9. Apply discount.
10. Generate final bill.
11. Exit.

Food Item Details

Maintain: Food ID, Food Name, Category, Price, and Available Quantity.

Required Validation

The program must handle: Duplicate Food ID; Food item not found; Invalid food quantity; Negative quantity; Quantity greater than available stock; Empty order; Invalid menu choice; Invalid price; and Empty food name.

Bill Calculation

The program should calculate the subtotal by adding $\text{Price} \times \text{Quantity}$ for all ordered items. Apply an appropriate discount based on the total amount and display the final bill in a meaningful format.

5. Task 3 – Vehicle Rental Management System

Develop a console-based **Vehicle Rental Management System**.

The system should allow the user to:

1. Add a vehicle.
2. Display all vehicles.
3. Search for a vehicle.
4. Rent a vehicle.
5. Return a vehicle.
6. Display available vehicles.
7. Display rented vehicles.
8. Exit.

Vehicle Details

Maintain: Vehicle ID, Vehicle Name, Vehicle Type, Rental Price Per Day, and Availability Status.

Required Validation

The program must handle: Duplicate Vehicle ID; Vehicle not found; Renting an already rented vehicle; Returning an available vehicle; Invalid Vehicle ID; Negative rental price; Empty vehicle name; and Invalid menu choice.

6. Testing Requirement

Prepare and execute at least **10 test cases**. The test cases must include both valid and invalid scenarios.

TC ID	Input / Scenario	Expected Result	Actual Result	PASS/FAIL
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TC-01	Valid record	Record added successfully		
TC-02	Duplicate ID	Error message		

TC ID	Input / Scenario	Expected Result	Actual Result	PASS/FAIL
TC-03	Invalid ID	Appropriate error		
TC-04	Invalid numerical input	Input rejected		
TC-05	Valid operation	Operation successful		
TC-06	Record not found	Error message		
TC-07	Boundary value	Correct result		
TC-08	Invalid menu option	Error message		
TC-09	Invalid/negative value	Input rejected		
TC-10	Valid transaction	Correct output		

Note: Use a menu-driven interface. Implement proper input validation and error handling. Students should demonstrate all major operations during the laboratory session. Individual implementation and understanding will be evaluated.